Technical Report - **Product specification**

PlantPatrol

Course:	IES - Introdução à Engenharia de Software
Date:	Aveiro, 4/10/2024
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Project abstract:	A platform to help manage greenhouses for a plant store, in an eco-friendlier way.

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1 Introduction

In the context of the final project for the IES course, our group came up with the idea of a project related to the management of data captured through real sensors, using microcontrollers and a Raspberry Pi 5, which was recently acquired by one of the group members. This further motivated us to use them, as it is an advanced topic that tests not only our personal skills, but also our ability to work as a team and organize task execution at a collective level.

As for the project, PlantPatrol, it is a system aimed at providing useful and easy-to-manage tools for a plant store, offering features such as inventory management and monitoring plant conditions such as temperature, humidity, light exposure, and UV radiation, among others.

Although the project in a first instance seems simple, there are several pieces that need to be put into place for all the personas that are going to use the system. Those pieces are the components of our architecture, such as: the backend that will be responsible to communicate the information to the client and receive the data from the sensors; the frontend that will showcase all the information about the previous spoken key functionalities.

All of this is very relevant for us, as future Software Engineers, because most of the software companies use an organized software development process that we'll use in the process of building this software project and furthermore, it helps us to achieve a better understanding and know-how about the software architectures and technologies.

2 Product concept

Vision statement

A plant shop is expanding land and one of the biggest issues that they're facing is related to managing the conditions of so many plants, and as consequence, some end up dying because of lack of water due to clogged or broken water pipes and others for insufficient ventilation or unknown reasons. With a plant shop this large, currently it's nearly impossible and resource-demanding for the employees to watch out for all the plants in the land. Furthermore, it's becoming much more difficult for the manager to keep track of the plant's stock and monitoring the greenhouses in general.

This system will be used mainly as a tool to help employees of a plant shop to manage plants' health and its watering and ventilation systems using several sensors such as temperature sensors, humidity sensors, air quality sensors, etc. It also helps to manage the stocks by having a catalog and inventory management system, which helps the greenhouse to never run out of seeds or plants of a certain type, for example. This catalog could also be available for the customers to see, helping them to choose a new plant, or simply viewing the characteristics of a plant that they already bought.

This system will also implement a mini-chat widget that allows bidirectional conversation between customers and employees, if a customer wants to make a special order or only has a doubt about a plant, he may use that chat to talk with some employee.

We also thought about making an e-commerce platform that allows the customers to order plants online, but after thinking more deeply, we decided that this requirement is a bit out-of-scope, so we decided not to go with it.

Overall, this system will allow better resource management which will lead to long-term cost savings and also provide a better customer experience for this growing business.

Personas and Scenarios

Personas

1. **João Ramalho**

   Age: 35 years

   Occupation: Greenhouse Business Owner and Manager

   Background: João opened his business when he was 23 years old. He oversees multiple greenhouses and a small team of employees.

   Motivation: He wants to make the store more efficient, by having automatic watering and monitor stock, conditions and employees all in one place.

2. **Paulo Miranda**

   Age: 26 years

   Occupation: Has been working for PlantPatrol for two years

   Background: Paulo is a skilled technician and his daily tasks include monitoring plant health, adjusting equipment (like sprinklers and heaters), cataloging and selling products.

   Motivation: Although Paulo likes what he does for work, he finds it would be easy if there was a centralized system where he could keep track of everything.

3. **Ana Martins**

   Age: 21 years

   Occupation: Student

   Background: Ana is a student who enjoys growing plants as a relaxing hobby. Taking care of her plants brings her joy and helps her unwind after long days.

   Motivation: While Ana is not an expert in horticulture, she loves learning more about plant care and is always on the lookout for new plants to add to his collection. She appreciates convenience, especially when it comes to getting new plants and keeping them healthy.

Principal Scenarios

Scenario 1:

One afternoon, while João is preparing for a meeting with suppliers, he receives an alert on his phone. The temperature in Greenhouse 2 has risen above the optimal range due to an unexpected heatwave. João immediately opens the PlantPatrol app, checks the live sensor data, and sees the spike in temperature. To prevent damage to his crops, he remotely adjusts the ventilation system through the app to increase airflow and cool down the greenhouse. He also schedules an automated ventilation increase for later in the day to ensure the temperature remains stable, all without needing to be on-site.

Scenario 2:

Paulo Miranda, a greenhouse technician, is responsible for providing customer support for plant care issues. One afternoon, Paulo receives a message through the PlantPatrol chat from Ana Martins, a regular client. Ana recently bought a fern, but she's concerned that its leaves are turning yellow. Paulo accesses the plant's profile in the system to review its care requirements and asks Ana a few questions about the conditions in her home, such as light exposure and watering frequency. After identifying that the issue might be due to overwatering, Paulo advises Ana to water the fern less frequently and move it to a spot with indirect sunlight. He logs the interaction in the system for future reference, ensuring that Ana is satisfied and that her plant will recover with the proper care.

Scenario 3:

Ana Martins, a plant enthusiast, often likes to browse the local greenhouse's selection before making a trip. One evening, she uses the PlantPatrol catalog to check the availability of a specific type of fern she's been wanting to add to her collection. The system shows that the fern is currently out of stock. Relieved that she checked beforehand, Ana saves herself a wasted trip to the store. She sets up an alert in the catalog so that she'll be notified the next time the fern becomes available. Now, Ana can plan her next visit to the greenhouse when the plants she's interested in are back in stock, saving both time and effort.

Product requirements (User stories)

User stories

1. As a Manager I want to control the plant's status so that I can see if everything is good, and issue an alert to the employees if something is wrong
    a. The system allows to see real-time data about the sensors
    b. The system allows you to issue an alert to employees about the care that should be taken for a given group

2. As a Manager I want to see historic statistics about what the sensors have recorded, so that I can see if the systems are well configured
    a. The system allows you to view graphs showing the atmospheric conditions for each group of plants
3. As a Manager I want to check up the inventory so that I can know what to order in advance
    a. The system allows you to define the minimum quantity of a product
    b. A notification is sent to inform you that the stock of a certain product is running low
4. As a Manager I want to add and manage the employees accounts on the system so that a new hired employee can access the system
    a. Authentication system
    b. The system allows you to change passwords
    c. The system allows you to define the responsibilities of each employee
5. As an Employee I want to receive a notification if a greenhouse sensor detect something abnormal so that I can adjust the systems
    a. Push notification system
6. As an Employee I want to setup the watering system based on sensor data so that I can manage the conditions easily and save water and energy
    a. The system allows you to define the conditions to which a group is subject to (amount of water supply, etc.)
    b. When the conditions have been changed, the user shall be notified with a status message.
7. As an Employee I want to update the stock of plant's so that the client and manager can access the current plants available
    a. The system allows you to change the quantity of each product
8. As a Employee I want to be able to help previous clients with their plant issues to that I have happy customers and a good plant support
    a. The system has a chat for a plant support line that allows the employees to speak and help customers.
9. As a Client I want to see the available plants on the store so that I can know when to buy and how many to buy
    a. The system allows you to check whether a given plant is available in a given greenhouse
    b. Notification system (for example by email) that notifies when a plant is available again
10. As a Client I want to be able to talk with the employees through a chat to that clarify my doubts about issues that may occur
    a. The system has a chat for a plant support line
11. As a Client I want to know which plants have certain characteristics so that I know which will do best in my garden.
    a. The system allows you to check the ideal characteristics for each plant

Epics

- Manage plant stock- 3,7
- Give support to the clients -8,10
- Notification System-1,3,5,6,9
- Sensor data system-1,2,5,6
- Employees permission and tasks system-1,4

3 Architecture notebook

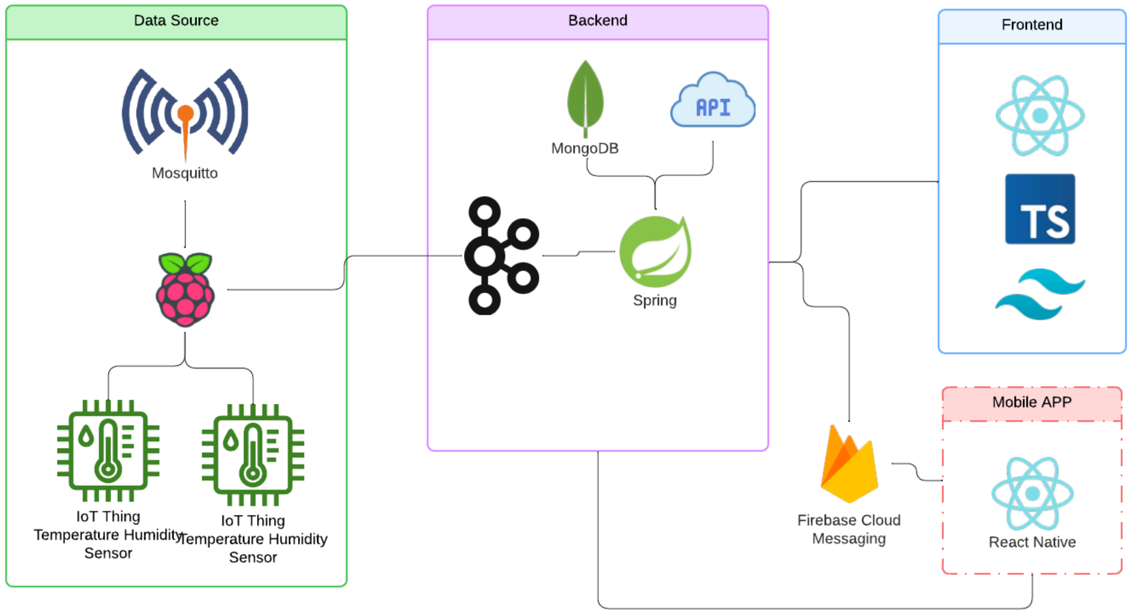
Key requirements and constraints

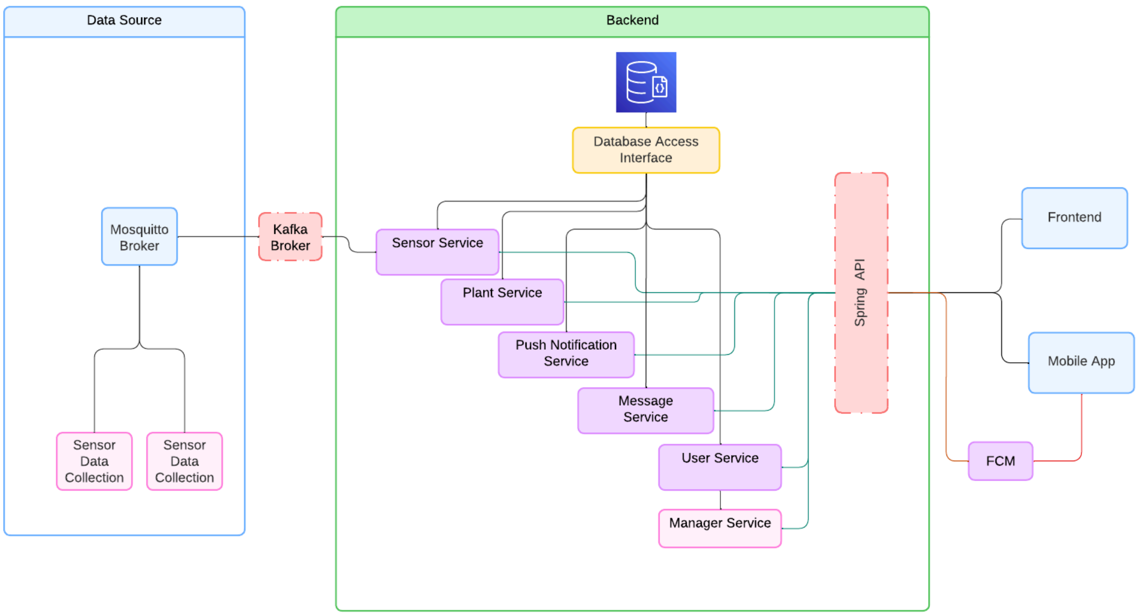
The architecture aims to provide great flexibility of components and their extensibility for this system. One of the main issues encountered was the connectivity between the various controllers, as well as the message flow they would follow to the database, without removing the system's partitioning or compromising the availability of the information provided by each component. The components in our context are, esp-8266 micro-controller, raspberry pi and the main server where the database and the API will be publish. There are some key requirements and system constraints that have a significant bearing on the architecture. They are:

- All sensor information shall be sent even if a between component is not available.
- If the backend, spring boot api is overloaded with requests the broker that accepts that sends the data from the sensors shall be able to retain persistent data if necessary
- The frontend requests have to be always able to make any type of request even if the data from the sensors are not fully updated by the second.
- The raspberry pi has to always send data on a set interval for a heartbeat connection to reassure the connection with the backend in case of a problem or lost connection between the parts.
- The connection between the micro-controllers and the raspberry pi, the mosquitto broker, must stay isolated from the backend. And the raspberry pi being the bridge between the two.
- Each micro-controller is responsible to send data to a mosquitto pub/sub topic with information about the humidity, temperatures, UV light and Air quality.

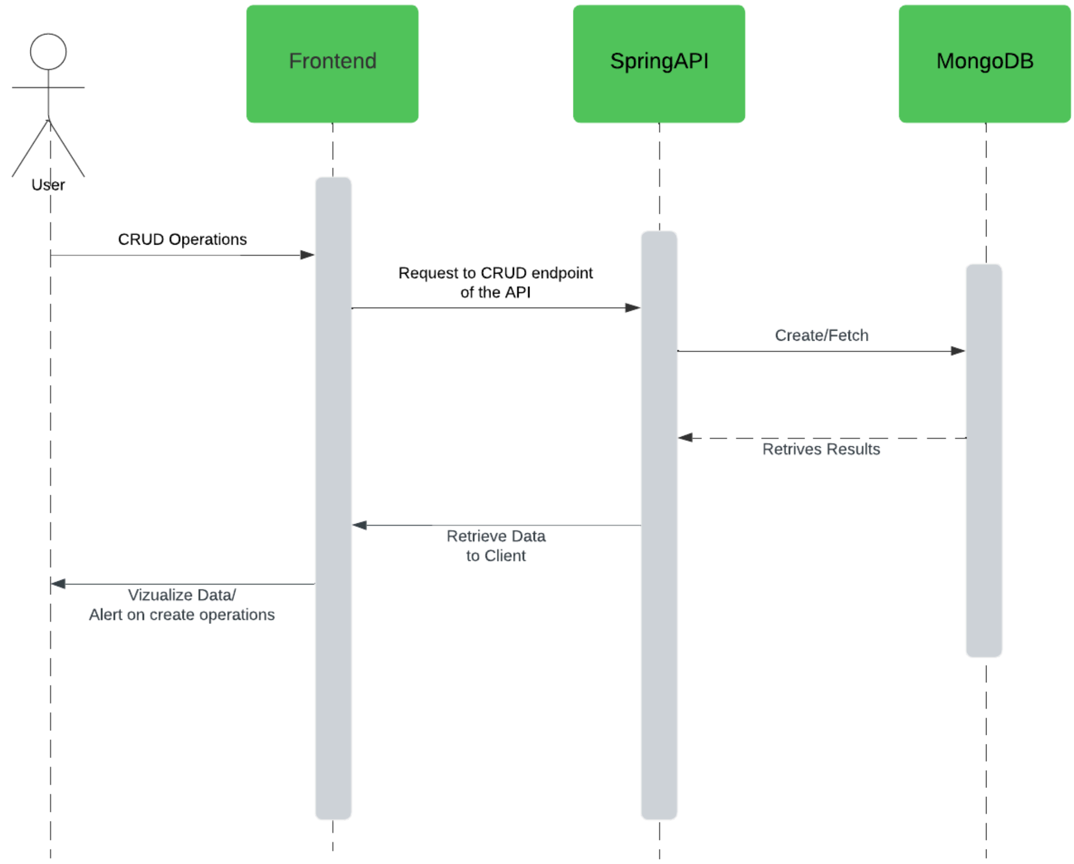
Architectural view

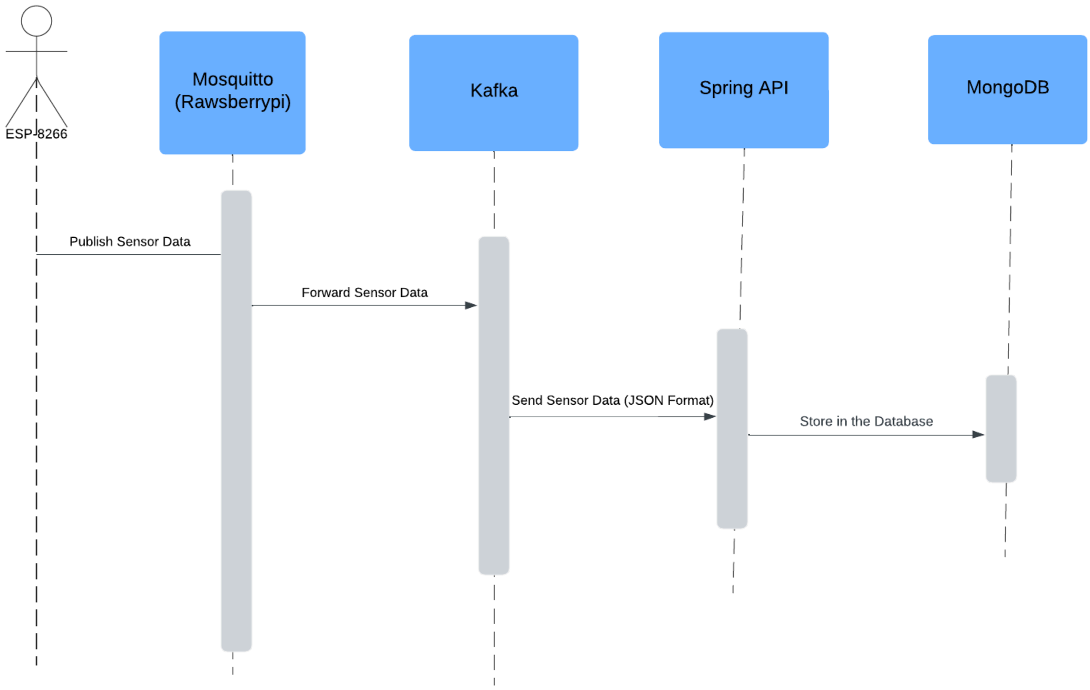
- [ ] **Data Source**: The data sources are divided into two main parts the micro-controllers and the Rawsberry PI, we also may develop “virtual” sensors that allow emitting custom data for testing.
- [ ] **Backend**: For the backend, we’ll use Spring Boot to implement a RESTful API and it’ll be the bridge between the frontends and the data layer, it will communicate with external APIs, for example to getting images and descriptions of plants.
- [ ] **Data Persistence Layer**: For the database, we plan to use MongoDB, as it is a document-based database, and our system’s data is not well-defined, and it is very volatile, so using a SQL-based database is not ideal for the system at hand.
- [ ] **Message Brokers**: The message broker will receive data from the virtual sensors/Raspberry Pi.
- [ ] **Web Frontend**: For the application UI, we’ll use React to make a website that will consume the REST API endpoints
- [ ] **Mobile app**: The mobile app will be used by the plant store clients, allowing them to view the plant's catalog and chat with the employees.
- [ ] **FCM (Firebase Cloud Messaging)**: We plan to use the Firebase Cloud Messaging as a push notification service, that allows the backend module to emit notifications that will be received by mobile/web clients.


Data Source
Mosquitto
IoT Thing
Temperature Humidity
Sensor
IoT Thing
Temperature Humidity
Sensor
Backend
MongoDB
API
Spring
Frontend
TS
Mobile APP
Firebase Cloud
Messaging
React Native


Data Source
Mosquitto
Broker
Sensor
Data
Collection
Sensor
Data
Collection
Kafka
Broker
Backend
Database Access
Interface
Sensor Service
Plant Service
Push Notification
Service
Message
Service
User Service
Manager Service
Spring API
Frontend
Mobile App
FCM

Module interactions





The mobile app on the frontend side is connected to FCM (Firebase Cloud Messaging) that is responsible to send data to the backend information about the push notifications, then the backend communicates externally with the FCM to send the notification itself.

We plan to use the Google custom search API to search and fetch images of plants when an employee registers a new plant in the catalog or inventory. We also plan to use an LLM API (for example OpenAI's gpt-3.5 API) to generate descriptions of the added plants.

4 Information perspective

<which concepts will be managed in this domain? How are they related?>

<use a logical model (UML classes) to explain the concepts of the domain and their attributes>

5 References and resources

- https://cedalo.com/blog/how-to-use-mqtt-on-esp8266/
- https://dev.to/nimrodresulta/install-kafa-to-raspberry-pi-4-43hp